Table I. Size characteristics of surface material (< 2 mm) on the runoff plot

Sample	% Sand*	% Silt*	Clay
1	66	23	10
2	76	15	9
3	87	8	5
4	88	8	4
Average	79·3	13·7	7·0

* Throughout this paper the limiting size between sand and silt is that of the British Standards Institution (i.e. 60 μm)

were installed which permitted the sampling of interrill overload flow with minimal flow. Details of the design and use of these flumes have been given elsewhere (Parsons and Eight such miniature flumes were installed along S1, and 10 along S2 (Figure 2). Along flumes were positioned so as to sample the range of overland flow depths. Flumes of three (10, 20, and 40 cm), the size of flume installed at any location being determined by the and flow discharge. The narrowest flumes were used where discharge was expected to be he fill times of the sample bottles would not be too short. Access to the miniature flumes on the plot surface was gained via small metal gratings supported about 10 cm above the arrow legs and placed just downslope of S1 and S2.

S2, part of the overland flow became concentrated into the rill, through which it reached a e located at the outlet of the plot (OF). The remaining interrill runoff was trapped in gutters red lower boundaries of the plot (Figure 2). Flow from these gutters was directed into (CF1 and CF2) to which automatic stage recorders were attached. However, as a result of raphy and the location of the boundary of the plot, much of the flow into the righthand e sufficiently concentrated as to create a rill. Consequently, the flow through CF2 cannot be interrill flow. Thus, only data from CF1 are comparable to those collected at S1 and S2 and n this study.

mulation experiments (E1, ..., E4) were performed on the plot lasting respectively for 10, 20, es, the intervals between successive experiments being 48, 1, and 72 hours. At the design hese four experiments correspond to rainstorms with a return period of 3, 5, 3, and 10 years, orn and Renard, 1988). During experiments flow passing over the lips of the miniature he calibrated flumes, and through the supercritical flume was sampled as often as possible, o to three minutes, by directing the water into one- or two-litre sample bottles. The samples niniature flumes and from CF1 are the basis of this study.

ESTIMATION OF SEDIMENT SIZE CHARACTERISTICS

during runoff events may be in the form of primary particles or aggregates. The f these particles is a function of the size distribution of the form in which they are eroded. ify this size distribution rather than that of the dispersed sediment has led some researchers ticle size analysis in the field immediately after samples have been collected from the runoff *t al.*, 1980). However, very sandy soils, such as those of the present study plot (Table I), are ated, and so tend to erode mostly as primary particles (Young, 1980). Under these laboratory determination of the dispersed sediment size distribution may be taken to be ld determination of the size distribution of the particles as they are eroded. In this study we ratory determination, as follows.

ontained within each sample of discharge was collected by allowing the samples to stand in r a minimum of 24 hours, decanting off most of the water, and then evaporating the